

## CM 8205 สื่อ เทคโนโลยีและวัฒนธรรม



Chutisant Kerdvibulvech, Ph.D.  
Associate Professor

รศ.ดร.ชุตินันต์ เกดวิบูลย์เวช

L1

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## INSTRUCTOR

Assoc. Prof. Dr. Chutisant Kerdvibulvech

รศ.ดร.ชุตินันต์ เกดวิบูลย์เวช

- Email: [chutisant.ker@nida.ac.th](mailto:chutisant.ker@nida.ac.th)
- Website: [www.gscm.nida.ac.th/~chutisant/](http://www.gscm.nida.ac.th/~chutisant/)
- Facebook: <https://www.facebook.com/chutisant.kerdvibulvech>
- Office: 8<sup>th</sup> Floor  
(Please Email/Facebook me first to make sure that I will be there.)

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## Course Syllabus and Lectures

For protecting our environment, all course syllabus and lectures can be downloaded online via the following links.

- **Course Syllabus**
  - <http://gscm.nida.ac.th/~chutisant/PhD/CS-CM8205.pdf>
- **Lectures**
  - <http://gscm.nida.ac.th/~chutisant/PhD/L1-CM8205.pdf>
  - <http://gscm.nida.ac.th/~chutisant/PhD/L2-CM8205.pdf>
  - etc

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## Evaluation (50%)

- Assignments, Participation and Attendance  
5 %
- Project  
45 %  
(Technique-based, Creativity-based, and Research-based)
- Extra Points  
(Rinkou)  
10 %

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## Textbooks

- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, ISBN 9786167687216
- ชุติสันต์ เกตวิบูลย์เวช. (2559) สื่อดิจิทัลใหม่..สื่อแห่งอนาคต. คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์.
- ชุติสันต์ เกตวิบูลย์เวช. (2560) พิมพ์ครั้งที่ 2, สื่อดิจิทัลใหม่..สื่อแห่งอนาคต. คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์.

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## Academic Papers (Scopus)

- -C. Kerdvibulvech, "Hand Tracking by Extending Distance Transform and Hand Model in Real-time," Pattern Recognition and Image Analysis, Vol. 25, Issue 3, pp. 437441, Springer Publisher (INSPEC, SCImago Q3, Elsevier's Scopus, EiCompendex, ACM, and Google Scholar), July 2015.
- -C. Kerdvibulvech, "A Study of Interactive Digital Multimedia Applications," The Pacific-Rim Conference on Multimedia (PCM'15), Lecture Notes in Computer Science (LNCS), part I, vol. 9314 (DBLP, SCImago Q2, and Scopus), Springer International Publishing Switzerland, Gwangju, Republic of Korea, pp. 192199, September 1619, 2015.

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## Websites and Links

- <https://www.scimagojr.com/>
- <https://www.sciencedirect.com/>
- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>

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## Overview

- Background
- From Digital Media to New Digital Media
- 3D Display / 3D Television
- Augmented Reality

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## Myself

- Received PhD in Computer Engineering from Keio University, Japan (2009)
- Assistant Professor at GSCM, NIDA (2015-Now)
- Head of Department of Information and Communication Technology, Rangsit University (2009-2014)

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## Academic Work Experience

- Invited Lecturer and Research Committee in many universities in Thailand, such as
  - Chulalongkorn University
  - Sripatum University
  - Panyapiwat Institute of Management
  - Mahidol University
  - Kasetsart University
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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# Media Work Experience

## Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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# Media and Technology



## Digital Media

Digital media are any media that are encoded in a machine-readable format. [*Technology Brief*. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



[www.linkedin.com](https://www.linkedin.com/)

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## Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

### Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.
2. Television and newspaper still matter, but it can no longer succeed as a stand alone platform.



::Source::  
Digital Media Trends and the Future of  
Television, Dr. Seth Geiger,  
President SmithGeiger LLC (CA, USA)

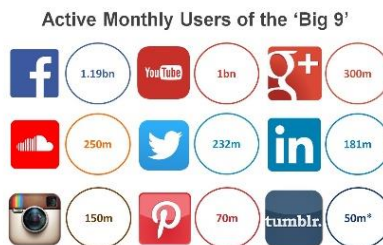
<https://www.linkedin.com/>

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## Digital Media Trends for 2014

### Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)  
\* Tumblr does not release active monthly user data and this figure is based on an estimate from Facebook's own data. The source is also a derivative of the 300m daily page views.

kamber

kamber

14 Social Media Trends for 2014  
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Kamber.com.au

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Source:  
 Kamber  
 (Sydney,  
 Australia)

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## Digital Media Trends for 2016

### facebook Community Update

4.27.2016



1.65 Billion  
 on Facebook each month



1 Billion  
 on WhatsApp each month



900 Million  
 on Messenger each month



400 Million  
 on Instagram each month

LIVE

Live Video  
 Launched worldwide



Connectivity  
 25 million connected  
 via Internet.org

Introduced Reactions



Artificial Intelligence  
 Access for the blind  
 and visually impaired



Oculus Rift  
 Shipped with 50+  
 games and apps

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<https://www.facebook.com/zuck>

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## Mobile Phone As Home Computer

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



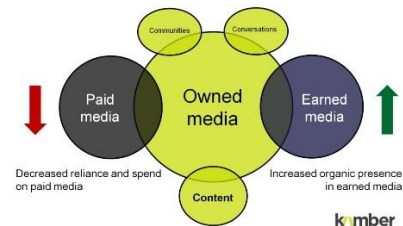
photos.suntimes.com

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## Paid Media / Earned Media

### The content marketing argument



kamber

kamber

14 Social Media Trends for 2014

Kamber.com.au

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::Source::  
Kamber  
(Sydney,  
Australia)

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## Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
  - Easy to build innovatively
  - Create HUGE profits

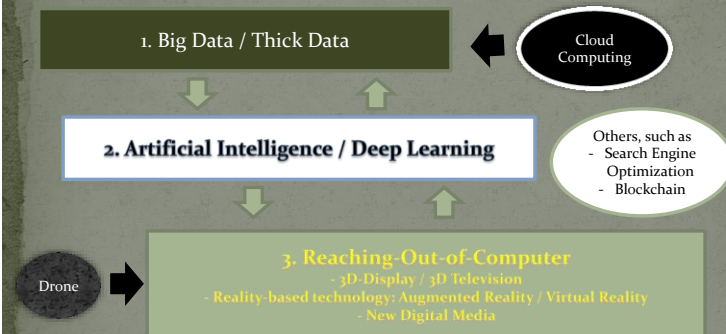


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<http://www.dailynews.co.th/>

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## The Future of the New Digital Age



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# Big Data

- “a field that treats of ways to analyze, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software”

Breur, Tom (July 2016). "Statistical Power Analysis and the contemporary 'crisis' in social sciences". *Journal of Marketing Analytics*, 4 (2-3): 61-65.

## Characteristics (5V of Big Data)

1. Volume
2. Variety
3. Velocity
4. Value
5. Veracity

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# Thick Data

Thick data is qualitative information that provides insights into the everyday emotional lives of consumers.

Tricha Wang, TED 2013  
<https://bigishpr.com/the-power-of-thick-data/>

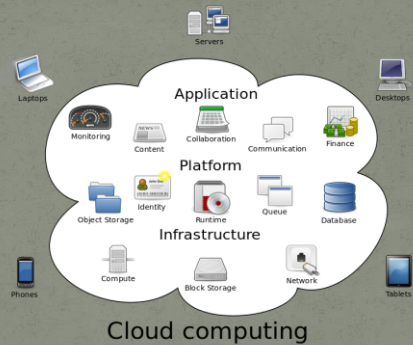
It goes beyond big data to explain why consumers have certain preferences, the reasons they behave the way they do, why certain trends stick and so on.

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# Cloud Computing

"...makes computer system resources, especially storage and computing power, available on demand without direct active management by the user."

[https://en.wikipedia.org/wiki/Cloud\\_computing](https://en.wikipedia.org/wiki/Cloud_computing)



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# Artificial Intelligence

- “..an area of computer science that emphasizes the creation of intelligent machines that work and react like humans..”

<https://www.techopedia.com/definition/190/artificial-intelligence-ai>

- Speech Recognition / Computer Vision
- Machine Learning / Deep Learning

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# 3D Display / 3D Television

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## 3D Display / 3D Television

The screenshot shows a news article from AFP. At the top, there's a navigation bar with the AFP logo and a search bar. Below the navigation bar, the article title is "Japan firm showcases Bat-Signal of the future". The date "20 OCT 2014" is displayed on the left. The main text describes a free-floating image created by firing lasers into thin air, unveiled in Japan on Monday. It mentions that the technology could be used to project messages into a cloudless sky, as seen in Batman. The article also notes that the company behind the technology, Burton Inc., showed off rotating spirals, fluttering butterflies and the outline of an apple hovering a few metres over a van loaded with the system. At the bottom of the article, the URL "http://www.afp.com/" is provided.

AFP

KIEV (AFP) - 11/11/2014 - 14:11  
< Risk of escalation in eastern Ukraine 'rising': OSCE >

THE NEWS THE AGENCY PRODUCTS AFP NEWS

Home:

### News

#### Japan firm showcases Bat-Signal of the future

20 OCT 2014

A free-floating image created by firing lasers into thin air was unveiled in Japan on Monday, offering the possibility one day of projecting messages into a cloudless sky, as seen in Batman.

And, like the Bat-Signal that alerted the Caped Crusader to trouble brewing in Gotham City, its inventors hope that it might eventually be used to warn people of danger such as an approaching tsunami.

The company behind the technology, Burton Inc., showed off rotating spirals, fluttering butterflies and the outline of an apple hovering a few metres over a van loaded with the system.

<http://www.afp.com/>

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## 3D Display / 3D Television

A free-floating image created by firing lasers into thin air



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<http://www.afp.com/>

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## 3D Display / 3D Television

2014 October

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo



2009

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<http://www.afp.com/>

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## 3D Display / 3D Television



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<http://www.afp.com/>

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## 3D Display / 3D Television

- Miraikan is located in Odaiba, Tokyo



[www.japan-guide.com](http://www.japan-guide.com)

[www.miraikan.jp/](http://www.miraikan.jp/) Assoc. Prof. Dr. Chutisant Kerdvibulvech

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## New Viewpoint Analysis

Prof. Takeo Kanade,  
Carnegie Mellon University (CMU),  
United States

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## Super Bowl

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

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## Super Bowl

- “Eye Vision”
- Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



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Eye Vision from Prof.  
Takeo Kanade, CMU

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## The Matrix



[www.imdb.com](http://www.imdb.com)

- “Virtualized Reality”
- People will be able to customize the perspective from which they watch, for example, from that of a particular object.



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New Viewpoint Analysis

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# Augmented Reality

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## What is Augmented Reality (AR)?



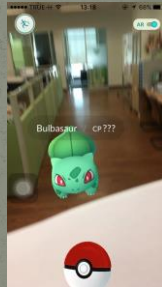
- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

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## What is the Goal of AR?

- To enhance a person's performance and perception of the world
- But, what is the ultimate goal?



*\*Augmented reality in urban places, connected content and  
data integration and analysis.*  
Assoc. Prof. Dr. Chutisant Kerdvibulvech  
Transactions of the Institute of British Geographers, 2012

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## The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce *Geek.com*  
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## Augmented Reality vs. Virtual Reality

### Augmented Reality

- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

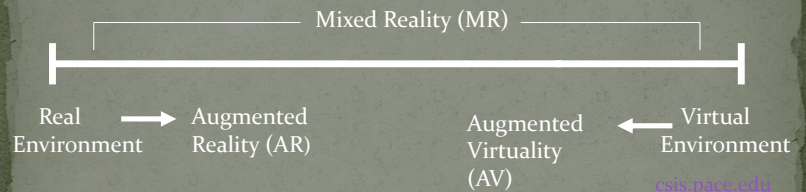
### Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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## Miligram's Reality-Virtuality Continuum



Milgram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer

generated data  
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## Combining the Real and Virtual Worlds

We need:

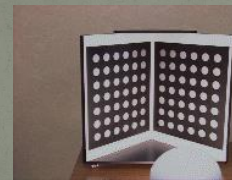
- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

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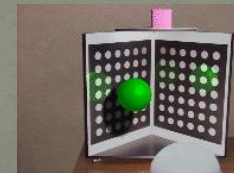
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## Combining the Real and Virtual Worlds (cont)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-  
reflections and virtual shading

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## Mixed Reality

- Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing

Journal: Personal and Ubiquitous Computing

2004



Dr. Adrian Cheok,  
National University of Singapore (NUS)  
at Singapore

Broadcast in Discovery Channel

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This image depicts and artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



[www.sngc3.co.jp](http://www.sngc3.co.jp)

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## Realistic Merging

Requires:

- Objects to behave in physically plausible manners when manipulated
- Occlusion
- Collision detection
- Shadows

\*\*All of this requires a very detailed description of the physical scene

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## Research Activities

- Develop methods to register the two distinct sets of images and keep them registered in real-time
  - New work in this area has started to use Computer Vision (CV) techniques
- Develop new display technologies for merging the two images

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## Performance Issues

Augmented Reality systems are expected:

- To run in real-time so that the user can move around freely in the environment
- Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- Update rate for generating the augmenting image
- Accuracy of the registration of the real and virtual image

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## Failures in Registration

Failures in registration due to:

- **Noise**
  - Position and pose of camera with respect to the real scene
  - Fluctuations of values while the system is running
- **Time delays**
  - In calculating the camera position
  - In calculating the correct alignment of the graphics camera

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## Applications

- Medical
- Military Training
- Hazard Detection
- Engineering Design
- Robotics and Telerobotics
- Training
- Sports
- Tourism
- Video conferencing

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## Applications (cont.)

- Entertainment, Computer Games
- Manufacturing, Maintenance, and Repair
- Consumer Design
- Audio
- Education
- Architecture
- Human welfare

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# Interactive Digital Multimedia Applications

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# Augmented Reality Applications

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# Sixth Sense

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## Sixth Sense

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

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## Sixth Sense in 21<sup>st</sup> Century



Dr. Pattie Maes,  
Massachusetts Institute of Technology  
(MIT), United States

Broadcast in TED

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## Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *“The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques”..from MIT Media Lab*

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## Detail of Sixth Sense

- The device recognizes items in stores, and then gathers and projects information about products to the user.
- It also is able to provide quick signals to let users know what the best products are based on their needs.”
- Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- Say goodbye to your encyclopedia

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## Cost of Sixth Sense

- Only \$350 dollars, around 12,000 baht
- From an educator’s stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- For educator’s, Sixth Sense would be a way to put technology (literally) into each student’s hands, because of its size and its cost.

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# Augmented Reality Guitar

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# Augmented Reality Guitar



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Keio University, Japan

Broadcast in Discovery Channel



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# Thank you

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Graduate School of Communication Arts and Management  
Innovation,  
National Institute of Development Administration (NIDA)  
[chutisant.ker@nida.ac.th](mailto:chutisant.ker@nida.ac.th)  
<https://www.facebook.com/chutisant.kerdvibulvech>

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